

OISD-STD-183

Standard on Logging Operations

Industry: Oil & Natural Gas

Category: Exploration & Production (Well Servicing / Wireline)

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1. STANDARD OVERVIEW

Standard Number: OISD-STD-183

Official Title: Standard on Logging Operations (Also referred to as Logging Operations Safety Standards)

Type: Standard (STD)

Industry Category: Exploration & Production (Upstream)

Main Subject: Comprehensive safety protocols for conducting well logging operations. The standard strongly emphasizes the handling, transportation, and field utilization of highly hazardous materials, specifically explosive charges and radioactive sources, deployed via wireline or tubing.

2. PURPOSE OF THE STANDARD

Well logging involves lowering sophisticated instruments into a wellbore to measure the physical, electrical, and radioactive properties of the rock formations. These operations routinely use highly potent radioactive sources (to measure porosity/density) and explosives (to perforate the steel casing or sever stuck pipes).

The primary purpose of **OISD-STD-183** is to establish stringent guidelines that prevent accidental detonations and radiation exposure. A unique aspect of this standard is its explicit focus on *human factors* in preventing accidents, recognizing that careless handling is the leading cause of catastrophic logging incidents.

3. SCOPE

The scope of OISD-STD-183 covers:

- **Wireline Well Logging:** Safe execution of open-hole and cased-hole logging.
- **Explosives Management:** Perforating operations (both wireline and tubing conveyed), gun loading at the base shop, and field deployment.
- **Radioactive Logging:** Operations involving density and neutron tools, and the safe storage, transportation, and handling of radioactive materials.
- **Logistics:** Safe transportation of equipment and personnel to the well-site.
- **Tool Interventions:** Use of wireline bridge plugs, packers, chemical/jet cutters, dump bailers, and pipe severing tools.
- **End of Life:** Proper disposal and liquidation of non-usable explosives and radioactive materials.

4. INDUSTRY APPLICATION

OISD-STD-183 applies throughout the drilling, completion, and production phases of the upstream lifecycle. It is the primary safety document utilized by:

- Wireline and Logging Service Contractors (e.g., Schlumberger, Halliburton, Baker Hughes).
- Wellsite Geologists and Petrophysicists.
- Base Shop Managers overseeing the loading of perforating guns and explosive magazines.
- Radiation Safety Officers (RSOs) attached to oilfield service companies.

5. MAJOR AREAS COVERED

The standard is structured around the lifecycle of the hazardous materials involved:

- Storage, Transportation & Safe Handling of Explosive Materials

- Gun Loading at Shop
- Pre-Departure Checks and Equipment Labeling
- Field Operations (Wellsite Safety Protocols)
- Radioactive Logging Operations
- Periodic Safety Audits
- Disposal of Hazardous Materials

6. DETAILED TOPIC-WISE SUMMARY

Gun Loading at Shop

What it covers: Preparation of explosive perforating guns at the service company's base before transport to the rig.

Important Requirements: The shop must have a designated, fixed gun loading area with limited explosive storage and strict security. Primary explosives (detonators) and secondary explosives (shaped charges) must be stored completely separately. Explosives must never be subjected to shock, blow, heat, or confining pressure. Warning signs like "EXPLOSIVES" and "NO SMOKING AREA" must be prominently displayed.

Pre-Departure Checks

What it covers: Ensuring all safety equipment is loaded before a logging truck heads to the field.

Important Requirements: All carrying cases containing explosives must have proper warning labels. The crew must load specific safety gear: a Safety Switch, Safety Tube, Explosive Remnant Box, casing-to-rig voltage monitor, a safety meter specifically designed for testing detonators, and a grounding device (C-Clamp) for the logging truck.

Field Operations (Wellsite Explosive Safety)

What it covers: Rigging up and firing tools containing electrically fired detonators.

Important Requirements: Electrically fired detonators are extremely sensitive to stray electrical currents and radio frequencies (RF). Mandatory warning boards must be displayed in Hindi, English, and regional languages: "*DANGER ZONE, EXPLOSIVE IN USE*", "*TURN OFF ALL RADIOS AND MOBILE PHONES*", and "*TURN OFF ALL GENERATORS*".

Radioactive Logging Operations

What it covers: Using tools emitting gamma rays or neutrons to log the well.

Important Requirements: Strict adherence to statutory regulations regarding storage and handling to prevent radiation exposure to the rig crew. Sources must be safely transported in heavy, shielded carrying cases and never handled by unauthorized personnel.

7. EQUIPMENT AND SYSTEMS

Equipment / System	Purpose	Standard Requirement / Context
Electrically Fired Detonators	Initiates explosive tools (perforation, severing).	Requires absolute isolation from stray voltage; rig generators and radios must be shut off.
Radioactive Sources	Used in density and neutron logging tools.	Must be transported in shielded containers and handled under strict radiation safety protocols.
Logging Truck Grounding System	Equalizes electrical potential between the truck and the rig.	Requires a C-Clamp and grounding cable to prevent static/stray currents from causing premature detonation.
Casing to Rig Voltage Monitor	Measures electrical potential difference.	Mandatory pre-departure equipment to ensure the environment is electrically safe before arming guns.

8. SAFETY REQUIREMENTS

- **Electrical / RF Safety:** Mandatory enforcement of "radio silence." The transmission of radio frequencies from walkie-talkies or cell phones can induce enough current in a wireline cable to prematurely set off an explosive detonator at the surface, causing a fatal explosion.
- **Explosive Management:** Scrap and unused explosive materials must be placed in non-sparking plastic containers (not metal cans or oil drums) and returned to the magazine for proper disposal/liquidation.
- **Human Factors:** The standard explicitly warns that unauthorized handling is the leading cause of accidents. Only trained, certified personnel may handle radioactive pills or detonators.

9. OPERATIONAL REQUIREMENTS

Before any explosive tool (side wall core guns, bridge plugs, chemical/jet cutters, pipe severing tools, dump bailers) is armed at the wellhead, the logging engineer must verify that the area is clear of non-essential personnel, warning signs are physically posted, and all stray voltage sources (generators, welding machines) are shut down and locked out.

10. INSPECTION, TESTING & MAINTENANCE

The standard mandates the use of specialized **safety meters** to test detonators. Standard multimeters are strictly forbidden, as their internal batteries can output enough current to trigger a detonator during testing. The standard also requires organizations to conduct "periodic safety audits" of their logging and explosive handling procedures.

11. EMERGENCY & HSE ASPECTS

A specific focus is placed on post-job housekeeping. The rig floor and loading areas must be meticulously cleaned after loading or unloading guns. Any explosive remnants must be secured in the "Explosive Remnant Box." There are also explicit requirements for the safe liquidation and disposal of expired or non-usable explosive and radioactive materials to prevent environmental contamination or theft.

12. IMPORTANT TECHNICAL CONCEPTS

Wireline Logging: The practice of lowering a specialized instrument package (a sonde) into a wellbore on the end of a continuously spooling electrical cable (wireline) to transmit data to the surface.

Primary vs. Secondary Explosives: *Primary explosives* (like detonators) are highly sensitive to heat, friction, and electricity and are used to trigger *Secondary explosives* (like RDX in shaped charges), which are more stable. OISD-183 mandates they are never stored or transported in the same compartment.

Stray Voltage: Unintended electrical voltage that exists between two objects (like the logging truck chassis and the rig casing) which must be neutralized via grounding before connecting detonators.

13. IMPORTANT TERMS & ABBREVIATIONS

Term / Abbreviation	Meaning	Relevance
RF Hazard	Radio Frequency Hazard	The risk of radios/phones inducing currents that set off detonators.

Term / Abbreviation	Meaning	Relevance
BHA	Bottom Hole Assembly	The tool string lowered into the well.
Dump Bailer	A wireline tool used to place cement/chemicals.	Often initiated by a small explosive charge, requiring OISD-183 compliance.
TCP	Tubing Conveyed Perforation	Explosives lowered on drill pipe/tubing instead of wireline; still governed by this standard.

14. RELATED STANDARDS & REGULATIONS

- **OISD-STD-185 (Wire Line Operations):** Works directly alongside OISD-183, focusing more on the mechanical aspects of wireline operations.
- **OISD-STD-191 (Oil Field Explosive Safety):** Provides broader, overarching rules for explosive magazines on oilfields.
- **PESO / CCE Rules:** Statutory Indian rules governing the magazines where the logging explosives are stored.
- **AERB (Atomic Energy Regulatory Board):** Statutory Indian authority governing the radioactive sources used in density/neutron logging tools.

15. PRACTICAL USE CASES

Scenario 1: Perforating a Production Casing

Relevant aspect: Field Operations (Electrical/RF Safety).

Why it matters: Before the wireline engineer arms the 40-foot perforating gun at the wellhead, the rig supervisor follows OISD-183 by enforcing total radio silence across the rig and turning off the main generator to eliminate the risk of a stray voltage detonating the gun on the rig floor.

Scenario 2: Preparing Tools at the Base Shop

Relevant aspect: Gun Loading at Shop & Storage.

Why it matters: A shop technician ensures that the detonators are stored in a separate bunker from the shaped charges, and ensures the loading area is designated as a strict "No Smoking" zone to prevent an accidental base explosion.

Scenario 3: Transporting Radioactive Logging Tools

Relevant aspect: Pre-Departure Checks & Transport.

Why it matters: The logging crew ensures the radioactive Cesium-137 source is locked in its heavy, lead-lined transport shield and correctly labeled before leaving the base, protecting the public and crew from radiation exposure during transport to the wellsite.

16. KEY TAKEAWAYS

1. OISD-STD-183 is the definitive safety standard for handling explosives and radioactive materials during well logging.
2. Human error and unauthorized handling are identified as the primary causes of logging accidents.
3. Primary explosives (detonators) and secondary explosives (charges) must be strictly isolated from each other.
4. Electrically fired detonators require total elimination of stray voltage and RF energy (radio silence) at the wellsite.
5. Pre-departure checklists must be rigorously followed to ensure grounding cables, safety meters, and warning signs are taken to the field.
6. Warning signs must be posted in English, Hindi, and local regional languages.
7. Only specialized safety meters can be used to test detonator continuity.
8. Proper disposal and liquidation of unused hazardous materials are mandatory to prevent theft or environmental harm.

17. INFORMATION LIMITATIONS

- **Verified Information:** The standard's scope, the specific list of explosive wireline tools covered, the exact wording required on field warning signboards, shop loading protocols, and pre-departure equipment lists.
- **Information Requiring Official Access:** Specific permissible radiation dose limits, the exact micro-ampere limits of the detonator safety meters, and the precise step-by-step procedures for the safe disposal of degraded explosives.

Note: "Detailed requirement is not publicly available in the sources reviewed." No quantitative radiation dosages or meter limits have been fabricated.

18. SOURCES & REFERENCES

1. **Scribd Document Repository** - Logging Operations Safety Standards (OISD-STD-183 extracts detailing Scope, Shop Loading, Pre-departure checks, and Field Operations).
2. **Oil Industry Safety Directorate (OISD)** - Official Standards and Guidelines List (Confirming Oct-2013 Edition in vogue).
3. **The Piping Engineering World** - List of OISD Standards.